A MORPHISM FROM AN OPEN SUBSCHEME INTO A PROPER SCHEME OVER THE BASE EXTENDS TO AN OPEN NEIGHBOURHOOD OF A VALUATION-THEORETIC POINT IN THE CLOSURE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Statement span 10424-10438 and proof span 10440-10473 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof contains an xymatrix square displaying the morphism from the spectrum of the local ring extending the one from the spectrum of its fraction field; it is copied verbatim and the retained chapter preamble loads Xy-pic.

Editorial note. The proof contains the author's parenthesis '(small detail omitted)' for the harmless replacement of the source by an open neighbourhood, and a reference to the author's earlier lemma producing a morphism from a prescribed germ; both are preserved.

Editorial note. 4 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The same-chapter valuative criterion of properness, in the form producing a morphism from the spectrum of a valuation ring compatible with a given morphism from its fraction field. Cited at proof lines 10443-10444. Produces the morphism on the spectrum of the local ring at the point, which is the candidate germ of the extension.
- The lemma producing an open neighbourhood and a morphism with prescribed germ, cited from the same chapter. Cited at proof lines 10449-10452. Spreads the germ at the point out to an actual morphism on a neighbourhood.
- The equalizer of two morphisms to a separated scheme over the base is a closed subscheme, cited from the schemes chapter. Cited at proof lines

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10455-10457. Shows that the two morphisms agree on a closed subscheme of the overlap, which contains the generic point.
- For every prime of a scheme there is an affine open containing it with the property that the ring of global functions injects into the local ring at that prime, cited from the properties chapter. Cited at proof lines 10427-10430. Lets the proof assume the source affine and integral with the local ring at the point a valuation ring whose fraction field is the residue field of the generic point.

Source reference. Original lemma 0BX5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14968–14968. Same-chapter reference used in the retained author text at source line(s) 10451. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 0BX6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14969–14969. Same-chapter reference used in the retained author text at source line(s) 10461, 12656. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 0BX3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14966–14966. Same-chapter reference used in the retained author text at source line(s) 10443. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01KM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2055–2055. Same-chapter reference used in the retained author text at source line(s) 1222, 10467, 12788, 12834. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let S be a scheme. Let X, Y be schemes over S. Let $x \in X$. Let $U \subset X$ be an open and let $f : U \to Y$ be a morphism over S. Assume*

(1) *x is in the closure of U,*
(2) *X is reduced with finitely many irreducible components or X is Noetherian,*
(3) *$\mathcal{O}_{X,x}$ is a valuation ring,*
(4) *$Y \to S$ is proper*

Then there exists an open $U \subset U' \subset X$ containing x and an S-morphism $f' : U' \to Y$ extending f.

Proof. It is harmless to replace X by an open neighbourhood of x in X (small detail omitted). By Properties, Lemma 0BX3 we may assume X is affine with $\Gamma(X, \mathcal{O}_X) \subset \mathcal{O}_{X,x}$. In particular X is integral with a unique generic point ξ whose residue field is the fraction field K of the valuation ring $\mathcal{O}_{X,x}$. Since x is in the closure of U we see that U is not empty, hence U contains ξ. Thus by the valuative criterion of properness (Lemma 0BX5) there is a morphism $t : \mathrm{Spec}(\mathcal{O}_{X,x}) \to Y$

fitting into a commutative diagram

$$\begin{array}{ccc} \mathrm{Spec}(K) & \longrightarrow & \mathrm{Spec}(\mathcal{O}_{X,x}) \\ {\scriptstyle \xi}\downarrow & & \downarrow{\scriptstyle t} \\ U & \xrightarrow{\quad f \quad} & Y \end{array}$$

of morphisms of schemes over S. Applying Lemma 0BX6 with $y = t(x)$ and $\varphi = t_x^\sharp$ we obtain an open neighbourhood $V \subset X$ of x and a morphism $g : V \to Y$ over S which sends x to y and such that $g_x^\sharp = t_x^\sharp$. As $Y \to S$ is separated, the equalizer E of $f|_{U\cap V}$ and $g|_{U\cap V}$ is a closed subscheme of $U \cap V$, see Schemes, Lemma 01KM. Since f and g determine the same morphism $\mathrm{Spec}(K) \to Y$ by construction we see that E contains the generic point of the integral scheme $U \cap V$. Hence $E = U \cap V$ and we conclude that f and g glue to a morphism $U' = U \cup V \to Y$ as desired. $\square$